

Science Topic 1 — Key-Point Notes 科學 Topic 1 重點筆記

Quiz 1 · Tue 6 Oct 2026 · from the real Topic 1 PowerPoints

1. Laboratory Safety 實驗室安全

- NEVER enter the lab unsupervised; do not touch equipment until told to.
- No running or horseplay; NEVER eat or drink in the lab.
- Wear safety goggles (plus gloves / apron when needed); tie back long hair, remove dangling jewellery, roll up sleeves.
- Report ALL accidents immediately, no matter how small.
- **Hazard symbols:** toxic, harmful, explosive, corrosive, highly flammable.
- **Safety equipment:** first aid kit, eyewash station, fire extinguisher / blanket.
- After the lab: wash hands with soap; NEVER touch, taste or smell chemicals unless told to; NEVER rinse anything down the sink without permission.

2. Measuring Accurately 準確量度

- Choose the right instrument — volume: cm^3 / ml / L · mass: g / kg · temperature: $^{\circ}\text{C}$ · time: s / min / hr · length: mm / cm / m / km.
- Measuring cylinder: stand it on a flat surface, eyes **level** with the liquid, read the **bottom of the meniscus**.
- Reliable results: repeat 3 times, keep the same decimal places, take the average.

3. The Bunsen Burner 本生燈

- **Parts:** Chimney (tall tube) · Collar (ring) · Air Hole (on the collar).
- **Lighting:** goggles on → burner on heat-proof mat → check rubber tubing on gas tap → CLOSE the air hole → light a small flame just above the chimney → turn on the gas.
- **Switching off:** close the air hole → turn off the gas → NEVER leave a flaming burner unattended.
- **Two flames:** air hole CLOSED = **luminous / quiet flame** (900–950 $^{\circ}\text{C}$) · air hole FULLY OPEN = **non-luminous / roaring flame** (1500–2000 $^{\circ}\text{C}$, hotter).

4. Scientific Method 科學方法

- **6 steps:** 1 Question → 2 Research / Observation → 3 Hypothesis → 4 Experiment → 5 Analysis → 6 Conclusion.
- **Hypothesis** = an educated guess, a possible answer to the question.
- **Qualitative** observation = senses (see, hear, feel, smell, taste), no numbers.
Quantitative = measurements and numbers.
- **Fair test** = shows how ONE variable affects another variable.

5. Variables, Accuracy & Graphs 變量、準確度及圖表

- **Independent variable** = what you CHANGE · **Dependent** = what you MEASURE · **Controlled** = kept the same (fair test).
- **Diagram rules:** pencil + ruler · at least 5 lines big · continuous strokes · label everything.
- **Accuracy** = close to the TRUE value · **Precision** = repeats close to EACH OTHER · **Reliability** = precision + repeatability (same person/equipment) + reproducibility (different person/equipment).
- **Graphs — 7 checks (H P L AL AS AU + pencil):** Heading, Plot accuracy, Line of best fit, Axis Label, Axis Scale, Axis Unit. Independent variable on the **x-axis**.
- **Line of best fit:** closest to MOST points — shows the trend, spots anomalous data. NEVER join dot-to-dot.